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Project Name	Auto Ticket Classification using Flow Designer

Project Documentation Format

(Adapted from FSD Documentation Format for Enterprise SaaS Architecture)

1. Introduction

Project Title: Auto Ticket Classification using Flow Designer

Team Members:

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2. Project Overview

Purpose: To automate the IT Service Management (ITSM) triage process. By replacing manual dispatching with a Machine Learning model integrated into workflow logic, the project aims to reduce Mean Time to Resolution (MTTR) and improve assignment accuracy.

Features:

- Natural Language Processing (NLP) to parse user-submitted incident descriptions.
- Automated, real-time ticket categorization and prioritization.
- Dynamic routing to resolver groups using ServiceNow Flow Designer.
- Fallback logic for low-confidence ML predictions.

3. Architecture

Frontend: The frontend utilizes the ServiceNow Service Portal, built on AngularJS, Bootstrap, and HTML/CSS, providing a seamless and responsive self-service experience for end users.

Backend: The backend logic is orchestrated entirely via ServiceNow Flow Designer (a low-code/no-code environment) and Server-Side JavaScript (Business Rules). The ML component uses ServiceNow Predictive Intelligence (powered by native ML clustering/classification algorithms).

Database: Relational cloud database (MariaDB) hosted on the ServiceNow PaaS. Interactions are handled via GlideRecord APIs, updating the primary `incident` and `sys_user` tables.

4. Setup Instructions

5. Folder / Component Structure

Because this is a cloud-native SaaS application, standard FSD directory structures translate into platform components:

Client (Frontend Components):

- `Service Portal (sp_portal)`: Main UI wrapper.
- `Widgets (sp_widget)`: Custom search and incident submission forms.
- `UI Scripts`: Client-side validation logic.

Server (Backend Components):

- `Flow Designer (sys_hub_flow)`: The visual workflow for ticket routing.
- `ML Solutions (ml_capability_definition)`: The classification models.
- `Business Rules (sys_script)`: Server-side data interceptors.

6. Running the Application

The application is hosted on the cloud. To access and run the application use the demo

7. API Documentation

The system exposes REST API endpoints for external integrations (e.g., third-party monitoring tools raising tickets):

- **Endpoint:** `POST /api/now/table/incident`
- **Description:** Creates a new incident and automatically triggers the ML classification flow.
- **Parameters:** `short_description` (String, required), `description` (String, optional), `caller_id` (String).
- **Example Response (201 Created):**

```
     { "result": { "sys_id": "8d6353eac0a8016400d8a125ca14fc1f", "number":
"INC0010023", "assignment_group": "Network Support" } }
```

8. Authentication

The platform utilizes Role-Based Access Control (RBAC). Authentication is handled via **OAuth 2.0** for API calls and Single Sign-On (SSO) through **Azure Active Directory** for internal employees logging into the Service Portal. GlideSystem session tokens manage active states.

9. User Interface

The User Interface is designed for simplicity. Users are presented with a "How can we help?" search bar and a generic ticket submission form. They only need to provide a title and a description. The complexity of categorization (e.g., selecting sub-categories, configuration items, or routing groups) is completely hidden from the UI and handled entirely by the backend ML.

10. Testing

Testing Strategy & Tools:

- **Automated Testing Framework (ATF):** Used to simulate user inputs, create dummy incidents, and assert that Flow Designer correctly updates the `assignment_group` field.
- **ML Validation:** The Predictive Intelligence model was tested using a 20% holdout validation set from historical data, achieving an 87% categorization accuracy.
- **UAT:** Manual User Acceptance Testing conducted by IT support leads to ensure routing logic aligns with business operational structures.

11. Screenshots or Demo

A full video demonstration of the ticket submission and automated routing process, along with the source Update Set XML files, can be found on the project repository:

Link: https://github.com/John-glacier/Auto-Ticket-Classification-using-Flow-Designer.git

12. Known Issues

- **Ambiguous Input:** Very brief or vague descriptions (e.g., "It is broken" or "Help") yield low confidence scores from the ML model. These fall back to a manual triage queue (L1 Dispatch).
- **Data Dependency:** The ML model requires periodic retraining (every 30-60 days) to recognize new IT vocabulary or new software systems introduced to the enterprise.

13. Future Enhancements

- **Virtual Agent Integration:** Integrating the ML model directly into a Chatbot/Virtual Agent so users receive real-time IT knowledge base articles before a ticket is even created.
- **Auto-Remediation:** Connecting specific high-confidence ML predictions directly to remediation scripts (e.g., automatically unlocking an AD account without human agent intervention).
- **Multi-lingual Support:** Training additional Natural Language models to support Spanish and French employee requests.